BioDINO-Edge: Biologically-Grounded Multi-Scale Vision Foundation Models for Trustworthy Few-Shot Plant Pathology on Resource-Constrained Hardware

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***Abstract*—Early and trustworthy diagnosis of plant foliar pathogens is pivotal for precision agriculture, global crop preservation, and automated robotic scouting. However, deploying contemporary vision models to agricultural edge devices faces two persistent bottlenecks: extreme data scarcity for novel emerging diseases (the low-data, few-shot regime) and tight hardware constraints (memory, latency, and thermal envelopes). In this work, we present BioDINO-Edge, an adaptive, biologically-grounded dual-foundation framework tailored for high-accuracy, real-time few-shot plant pathology on resource-constrained hardware.**

Rather than relying on generic internet-trained vision backbones that struggle with subtle histological chlorosis, BioDINO-Edge synergistically fuses two complementary foundation models: (i) a lightweight self-supervised visual patch extractor (DINOv2 Small, 22.1M parameters) serving as an ultra-fast local lesion feature encoder (23.4 ms on CPU), and (ii) a domain-specific biological vision-language model (BioCLIP ViT-B/16) pre-trained across 10 million organismal specimens to capture fine-grained taxonomic and pathological hierarchies. An adaptive uncertainty gate—formulated via normalized predictive Shannon entropy and top-margin confidence—dynamically orchestrates inference: high-confidence queries execute instantaneously along the fast edge path, whereas ambiguous or fine-grained foliar lesions trigger deep taxonomic consensus reasoning. Extensive evaluations on the PlantVillage 5-shot intra-species benchmark demonstrate that BioDINO-Edge achieves a record-breaking 94.70% Accuracy and 94.80% Macro F1-score, substantially outperforming generic CLIP PEFT baselines (88.02%) and resolving historical diagnostic bottlenecks such as Apple Scab (improving recall from 56.0% to 90.2%, F1: 94.0%) with 0.00% cross-species confusion and 0.00% false negatives on healthy grape foliage. Dynamic INT8 quantization demonstrates real-time edge execution (> 35 weighted FPS), providing an ideal paradigm for battery-powered agricultural edge robotics.

***Index Terms*—Plant Pathology, Few-Shot Learning, Vision Foundation Models, BioCLIP, DINOv2, Adaptive Uncertainty Gating, Edge AI, Precision Agriculture.**

I. Introduction

Global agricultural systems face unprecedented pressures from population growth and escalating crop disease outbreaks, which destroy up to 40% of agricultural yield annually. Automated early diagnosis via embedded computer vision deployed on uncrewed aerial vehicles (UAVs) and field-edge sensors is critical for localized fungicide micro-spraying and sustainable crop health management.

Despite rapid advancements in deep learning, real-world field deployment suffers from severe limitations:

1) **The Novel Pathogen Scarcity Bottleneck:** Collecting thousands of annotated agricultural images for emerging disease mutations or rare regional pathogens is economically infeasible. Edge systems must reliably diagnose infections given only $K = 5$ support shots.
2) **Representation Failure of Generic Foundation Models:** Off-the-shelf vision-language models (e.g., standard OpenAI CLIP [1]) are pre-trained on internet-scale multimodal data containing macroscopic consumer concepts (vehicles, pets, urban scenes). Consequently, they frequently exhibit representation collapse when distinguishing fine-grained leaf pathology, such as olive-green velvety *Venturia inaequalis* (Apple Scab) versus healthy foliage, leading to high false-negative rates.
3) **Edge Computational Constraints:** Agricultural sensors and handheld diagnostic devices operate on low-power ARM/x86 microprocessors with limited RAM (1–2 GB) and restricted battery budgets, rendering oversized multi-billion-parameter foundation models unusable.

To simultaneously resolve these representation and computational dilemmas, we propose **BioDINO-Edge**. Our core insight is that visual pathology diagnosis naturally aligns with a dual-process cognitive architecture:

- **Fast Spatial Feature Extraction (System 1):** Handled by a 22.1M-parameter DINOv2 Small backbone [2], which leverages self-supervised patch tokens to detect spatial textural anomalies at 42.7 FPS on edge CPU.
- **Deep Taxonomic Multimodal Reasoning (System 2):** Handled by BioCLIP [3], pre-trained on TreeOfLife-10M, aligning Latin botanical taxonomy (*Malus domestica*, *Vitis vinifera*) and organismal pathogen priors with visual features.
- **Predictive Uncertainty-Gated Cascading:** High-confidence foliar samples bypass secondary computation, while borderline, ambiguous lesions are dynamically routed to multimodal consensus.

II. Proposed BioDINO-Edge Architecture

A. Foundation Backbones Formulation

Given an incoming query leaf image $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$:

- The fast-path feature $\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$ is extracted via DINOv2 Small ($f_{\text{dino}}(\cdot)$) and ℓ_2-normalized:

$$\mathbf{f}_{\text{dino}} = \frac{f_{\text{dino}}(\mathbf{x})}{\|f_{\text{dino}}(\mathbf{x})\|_2} \tag{1}$$

- The domain-specific biological feature $\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$ is extracted via the BioCLIP visual transformer ($f_{\text{bio}}(\cdot)$) and ℓ_2-normalized:

$$\mathbf{f}_{\text{bio}} = \frac{f_{\text{bio}}(\mathbf{x})}{\|f_{\text{bio}}(\mathbf{x})\|_2} \tag{2}$$

B. Multi-View Prototype Synthesis

For novel classes $\mathcal{Y}_{\text{novel}} = \{1, \dots, C\}$ represented by support sets $\mathcal{S}_c$ with $K = 5$ images per class, we generate an 8-fold augmented multi-view support manifold $\tilde{\mathcal{S}}_c = \bigcup_{i=1}^{K} \mathcal{T}_{\text{aug}}(x_i^{(c)})$ incorporating random affine shear, color jitter, flip, and zoom transforms. Empirical centroid visual prototypes are computed as:

$$\mathbf{P}_{\text{dino}}^{(c)} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{x}\in\tilde{\mathcal{S}}_c} f_{\text{dino}}(\tilde{x})\right) \tag{3}$$

$$\mathbf{P}_{\text{bio, vis}}^{(c)} = \text{Normalize}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{x}\in\tilde{\mathcal{S}}_c} f_{\text{bio}}(\tilde{x})\right) \tag{4}$$

C. Taxonomic & Pathological Language Priors

To inject rigorous botanical grounding, we formulate scientific diagnostic descriptors utilizing Latin nomenclature and symptomatic morphology:

$$T_c = \text{“A botanical photo of } \mathcal{H}_{\text{latin}}^{(c)} \text{ exhibiting } \mathcal{D}^{(c)} \text{ caused by } \mathcal{P}^{(c)} \dots \text{”} \tag{5}$$

The BioCLIP text encoder $\phi_{\text{bio}}(\cdot)$ constructs language prototypes $\mathbf{P}_{\text{text}}^{(c)} = \text{Normalize}(\phi_{\text{bio}}(T_c))$, which are blended into multimodal prototypes:

$$\mathbf{P}_{\text{bio}}^{(c)} = \text{Normalize}\left((1-\lambda)\mathbf{P}_{\text{bio, vis}}^{(c)} + \lambda\mathbf{P}_{\text{text}}^{(c)}\right), \quad \lambda = 0.15 \tag{6}$$

D. Adaptive Uncertainty-Gated Edge Routing

Fast-path logits are computed via scaled cosine similarity: $\mathbf{z}_{\text{dino}} = \gamma_{\text{dino}}(\mathbf{f}_{\text{dino}}\mathbf{P}_{\text{dino}}^{T})$ with $\gamma_{\text{dino}} = 20.0$. Softmax probabilities $\mathbf{p} = \text{softmax}(\mathbf{z}_{\text{dino}})$ yield normalized predictive entropy and margin confidence:

$$\mathcal{H}_{\text{norm}}(\mathbf{x}) = -\frac{1}{\log C}\sum_{c=1}^{C} p_c \log(p_c + \epsilon) \tag{7}$$

$$\mathcal{C}(\mathbf{x}) = (p_{(1)} - p_{(2)}) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}))_{+} \tag{8}$$

If $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$, the query exits immediately with zero secondary overhead. Otherwise, the sample triggers BioCLIP and produces consensus logits:

$$\mathbf{z}_{\text{fused}} = w_{\text{bio}}\mathbf{z}_{\text{bio}} + (1 - w_{\text{bio}})\mathbf{z}_{\text{dino}}, \quad w_{\text{bio}} = 0.65 \tag{9}$$

III. Experimental Evaluation

A. Experimental Setup

We conduct rigorous few-shot evaluations on the standard PlantVillage pathology benchmark across 8 novel intra-species categories (Apple and Grape taxa: 40 support images, 7,193 test query images). All tests are evaluated under identical 5-shot seed partitions. Hardware latency is benchmarked on a standard 4-thread CPU under dynamic INT8 quantization.

TABLE I
Comparative 5-Shot Performance on PlantVillage

Method / Architecture	Accuracy (%)	Macro F1 (%)
Zero-Shot OpenAI CLIP [1]	23.45	21.38
Vanilla Tip-Adapter [4]	81.67	78.40
Generic CLIP + LoRA + Gate	88.02	85.09
Fast-Path (DINOv2 Small Alone)	88.98	89.08
Deep-Path (BioCLIP Multimodal Alone)	94.11	94.00
BioDINO-Edge (Proposed)	**94.70**	**94.80**

B. Diagnostic Class-Wise Breakdown

As detailed in Table II, BioDINO-Edge completely overcomes the representation plateau of generic models. Apple Scab, which suffered from a 44% false-negative rate in generic CLIP (recall: 56.0%), achieves **90.2% Recall** and **98.1% Precision** (F1: 94.0%). Cross-species confusion between Apple and Grape is **0.00%**. Furthermore, Grape Healthy leaves achieve **100.00% Recall** with zero false negatives and only 0.10% false positives.

TABLE II
BioDINO-Edge Per-Class Pathology Diagnostic Metrics

Class Name	Precision	Recall	F1-Score	Support
Apple Scab	0.981	0.902	0.940	625
Apple Black Rot	0.939	0.932	0.936	616
Apple Cedar Rust	0.888	0.996	0.939	270
Apple Healthy	0.977	0.990	0.983	1640
Grape Black Rot	0.945	0.835	0.887	1175
Grape Esca	0.871	0.960	0.913	1378
Grape Leaf Blight	1.000	0.990	0.995	1071
Grape Healthy	0.984	1.000	0.992	418
Macro Average	**0.948**	**0.951**	**0.948**	**7193**
Overall Accuracy		**94.70%**		

C. Edge Hardware & Real-Time Quantization Profiling

Table III summarizes edge deployment characteristics on CPU under INT8 quantization. DINOv2 Small executes in only 23.45 ms (42.7 FPS), requiring just 22.1 MB of memory footprint. The cascaded pipeline dynamically balances computation, yielding an ultra-competitive real-time profile suitable for field robotics.

IV. Conclusion

We introduced BioDINO-Edge, a foundational dual-vision architecture uniting self-supervised spatial patch modeling (DINOv2) with organismal taxonomic language supervision (BioCLIP). Guided by predictive uncertainty gating, BioDINO-Edge sets a new state of the art in few-shot plant disease

TABLE III
EDGE HARDWARE & INT8 PROFILING (4-THREAD CPU)

Configuration	Params	INT8 (MB)	Lat. (ms)	FPS
DINOv2 Small (Fast Path)	22.1M	22.1	23.45	42.7
BioCLIP ViT-B (Deep Path)	86.6M	88.5	69.98	14.3
BioDINO-Edge (Weighted)	**Adaptive**	**110.6**	**27.60**	**>35.0**

diagnosis (94.70% Accuracy, 94.80% F1) while ensuring low-latency, energy-efficient deployment on agricultural edge microprocessors.

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